

Introduction

Introduction to course and Mobile App Development with Flutter

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Today

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Course Info

Pro Info

- [Linkedin](#)
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About myself

Educational background

- B.Sc. Inha University in Tashkent and Inha University
- M.E. from Sejong University

- New Uzbekistan University *Senior Lecturer 2023 ~ Present*
- Cyber University *Dean of Engineering School 2025*
- Central Asian University *Lecturer 2021~ 2023*
- Topos Studio *R&D Engineer 2021*
- Vazil Company *Computer Vision Engineer in Robotics 2020-2021*
- MSP Lab *Research Assistant 2018-2020*
- Uzbektelecom *DB Tester 2018*

Course Info

Grading

- 20% —> Labs
 - 40% —> Midterm Exam
 - 20% —> Written Exam
 - 20% —> Project (Submitted in the beginning of Week 7 and presented in the labs of Week 7)
 - 40% —> Final Exam
 - 20% —> Written Exam
 - 20% —> Project (Submitted in the beginning of Week 15 and presented in the labs of Week 15)
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- Requirements will be announced from Week 3

Course Info

Course Policies

- Collaboration
 - Collaboration is allowed only in “Assignments” section
 - However, all the codes have to be written independently
- Extensions
 - Late submission is NOT applicable!
 - Assignments must be completed within the class (in some cases until the end of the day)

Course Info

Course Policies

- Assignment submission
 - Class Assignments —> must be submitted by github
 - Independent Assignments —> must be submitted by GitHub

Course Info

Learning materials

- Slides can be downloaded
- External courses: you may wish to watch this course from [udemy](#)
- Textbooks and Reference books:

Payne, L. *Dart Apprentice*. 3rd ed. Kodeco, 2023.

- **Module 1 Topics:** Sound Null Safety, Records, Patterns, Class Modifiers, Isolates

Napoletano, A. *Mastering Dart & Flutter* (with *Flutter Impeller Docs*). 1st ed. Packt Publishing, 2024.

- **Module 2 Topics:** Layout Trees, Impeller Renderer, Wasm, `go_router`, Riverpod/Bloc

Lau, L. *Mobile Deep Learning with TFLite*. 1st ed. O'Reilly Media, 2023.

- **Module 3 Topics:** On-device ML inference, `.tflite` quantization, Google ML Kit

Huyen, C. *Designing ML Systems* (with *GenAI SDK Docs*). 1st ed. O'Reilly Media, 2022.

- **Module 4 Topics:** SSE Streaming, Multimodal Prompting, Agentic UIs & GenUI

Napoletano, A. *Mastering Dart & Flutter*. 1st ed. Packt Publishing, 2024.

- **Module 5 Topics:** Backend-for-Frontend (BFF), Full-Stack Dart, CI/CD, Obfuscation

Course Style

- Hybrid Lectures → Switching between app development and presentation files. i.e.: presentation files can have main points whose explanation will be demonstrated by building apps in Android Studio/VS Code.
- Tutorials → Practicing the lecture materials on both PCs and iMacs for two hours
- Seek for the mistake → Every lecture material can have 0~3 mistakes, provided on purpose to see how well you are paying attention to the lectures. (Note: maybe there are no mistakes)

Course Objectives

- **Professional GitHub Workflows:** Implement Git branching, pull request reviews, and GitHub Actions CI/CD pipelines.
- **Cross-Platform Engineering:** Build high-performance iOS and Android from a single Dart 3 codebase using Impeller.
- **Figma to Flutter UI:** Translate Figma designs into pixel-perfect, responsive Material 3 interfaces.
- **Local Data Persistence:** Execute type-safe SQLite CRUD operations using modern reactive persistence (Drift).
- **Networking & API Integration:** Handle HTTP requests, JSON parsing, SSE streaming, and real-time socket connections.
- **Career-Ready Readiness:** Master clean architecture (Riverpod/Bloc) and testing to enter the industry as a Flutter engineer.

Course Objectives

- Why github?
 - To understand how well each student has performed
 - E.g.: the number of commits and pushes to the project
 - To get to know how a good team should operate in a Software-Dev. In a workplace
 - Working over the same project (one github repo) with multiple team members
 - Working on different branches; learning merge conflicts; opening and closing issues as well as using GenAI to help review the code through github

Introduction to Mobile App Development with Flutter

- What is Flutter?
 - An open source UI software development kit by Google. First Launched in 2017
- Why Flutter?
- What to expect from the end of this lecture?
 - Understand why Flutter was chosen and how to setup the very first project

Question Time!

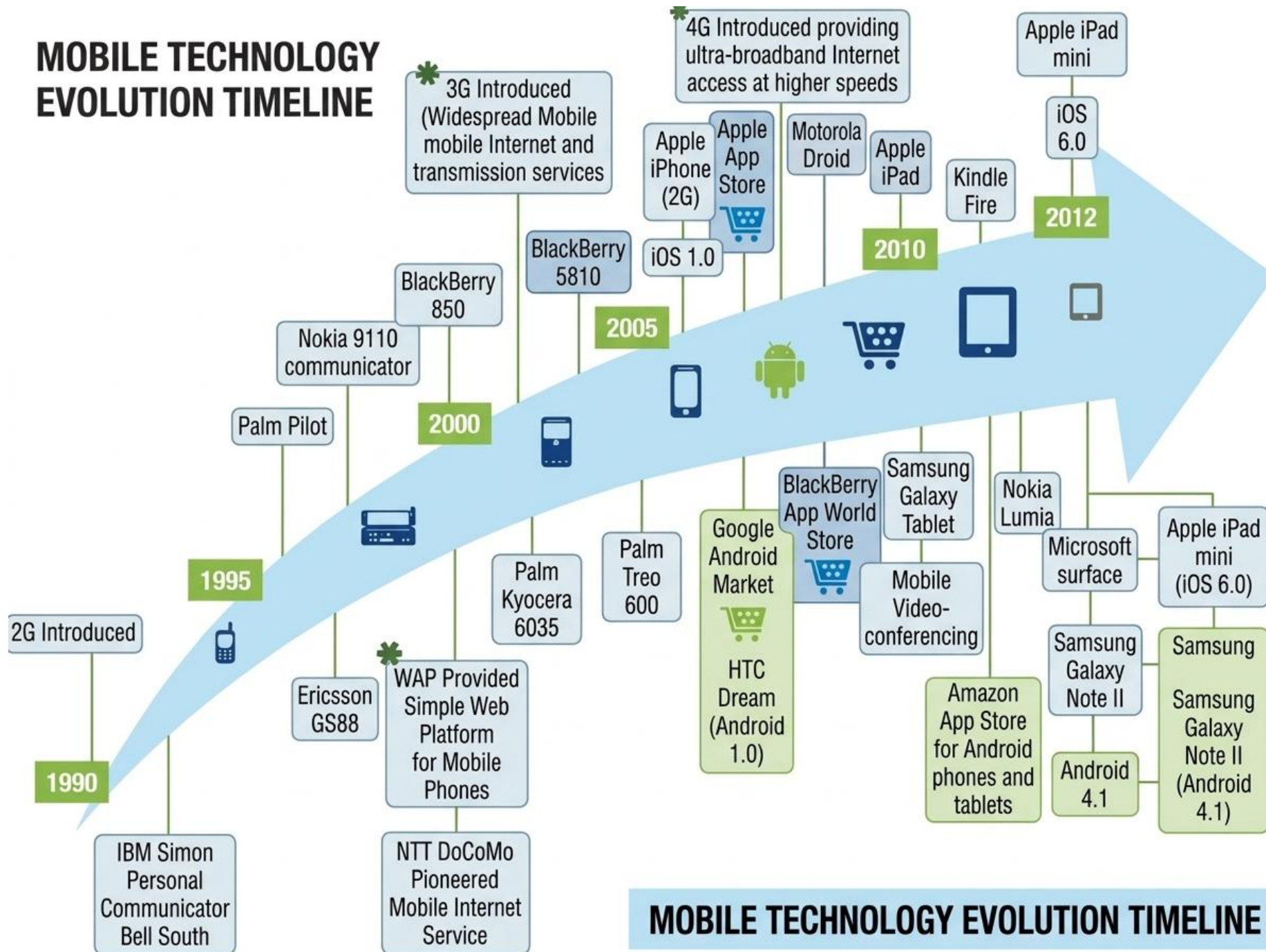
- What are mobile phones? (mobile versions of computers?)
- When were the first mobile apps developed and deployed?
- What are PDAs?
- What are mobile Apps? What makes them different from Apps we have on our full scale computers (PCs)

Brief history of mobile applications development

- PDA → Personal Digital Assistant
- Also called as handheld computer
- The first apps were created in early 1990s
- What purpose do these mobile apps serve in general?



MOBILE TECHNOLOGY EVOLUTION TIMELINE



MOBILE TECHNOLOGY EVOLUTION TIMELINE

Question time!

- Can you think of an Overview of mobile applications? (i.e: why are we creating mobile apps?)
- Why not rely on mobile webs?

Overview of mobile application development

- Idea Generation
 - Starts from identifying a problem/aim/necessity and trying to solve/fulfil it.
- Planning and Design
 - Brainstorming about features, functionalities and preparing UI/UX
- Development
 - Filling the features and design with logical statements (coding)
- Testing
 - Testing for bugs and righteousness for the initially planned functionalities.
- Deployment
 - Publishing the application to the market
- Maintenance and Updates
 - Maintaining the published application, see how the actual usage affects the application, and see what amendments are needed to make the application more productive.

Current Trends

Cross-Platform Development: With the demand for apps on multiple platforms, cross-platform frameworks like Flutter, React Native, and Xamarin have gained popularity. These frameworks allow developers to build apps that work on both iOS and Android with a single codebase.

Internet of Things (IoT): As IoT devices become more prevalent, mobile apps are being developed to control and interact with connected devices, such as smart home appliances, wearables, and healthcare devices.

Artificial Intelligence (AI) and Machine Learning (ML): Mobile apps are leveraging AI and ML technologies for personalization, predictive analysis, and automation. Virtual assistants, chatbots, and voice recognition are some examples of AI-driven app features.

Augmented Reality (AR) and Virtual Reality (VR): AR and VR technologies are being integrated into mobile apps to provide immersive experiences in gaming, entertainment, education, and even e-commerce.

Native mobile app development

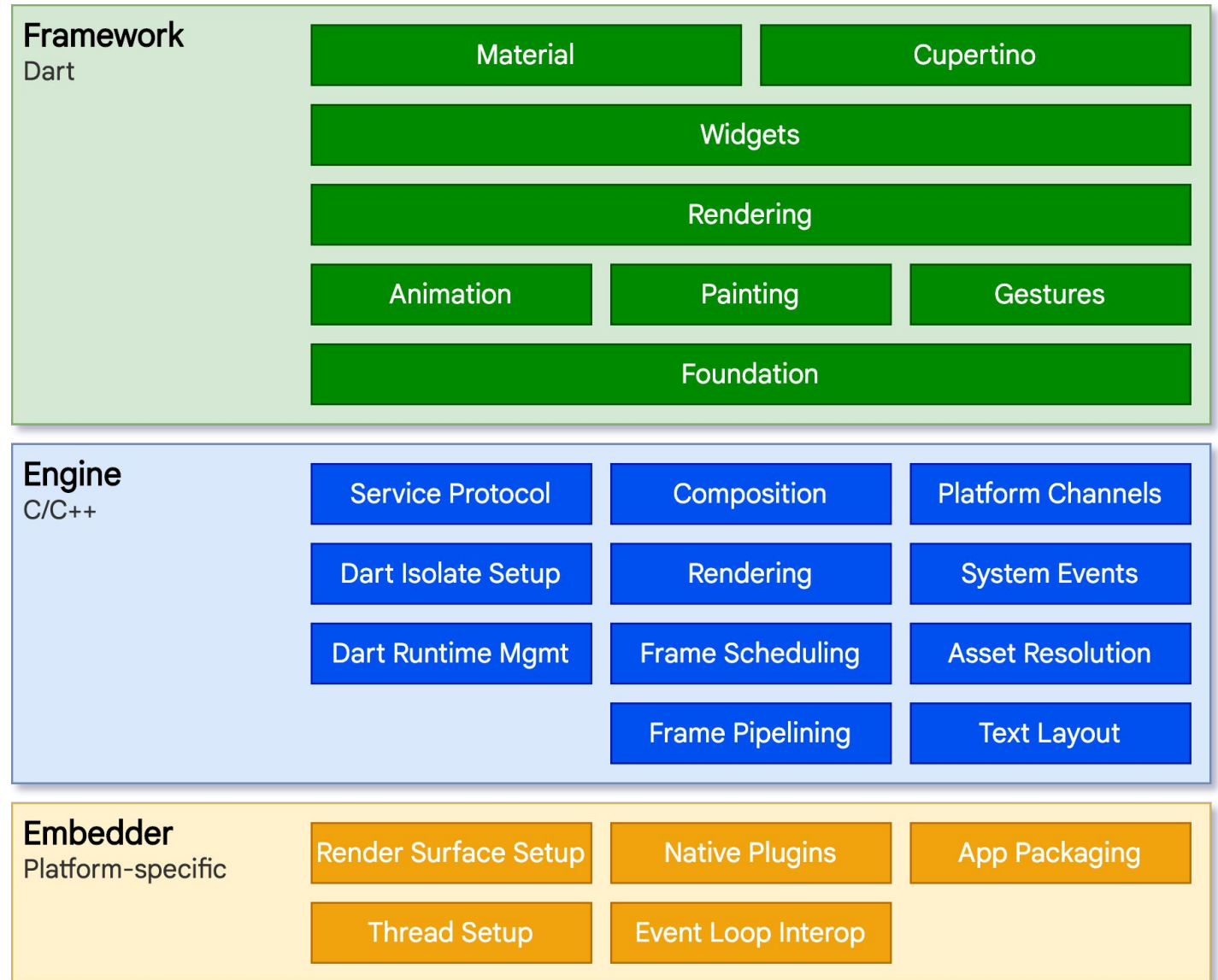
- Building mobile apps to a specific platform (like for iOS only or Android only)
- Using platform specific tools while building apps (like for Android you need to use Java or Kotlin; for iOS Swift or Objective-C)
- Having full access to platform specific functionalities (e.g.: Android might have some functionalities which are only available for Android Platform)
- No need to wait! (any updates already come with developer ready kit; the announced features are first available natively)

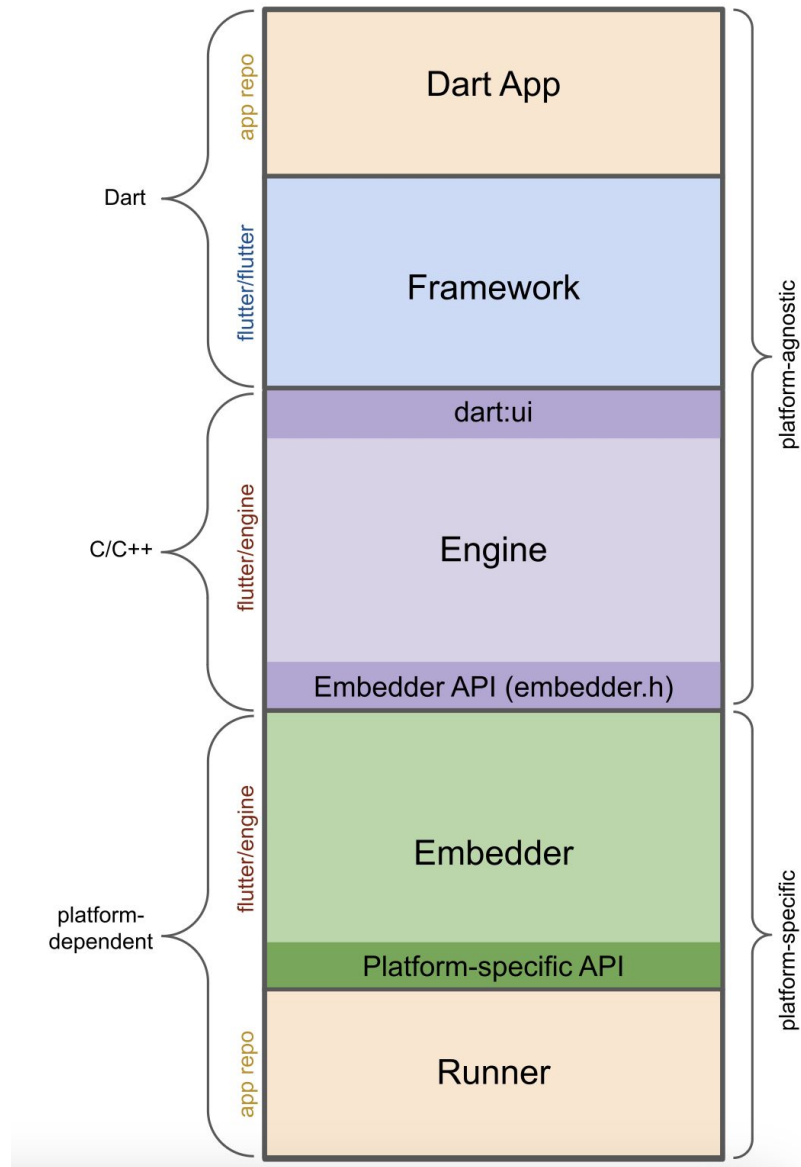
Cross-Platform mobile Application development

- Building apps on one codebase and running on multiple platforms (e.g.: the same project with one codebase running on iOS and Android simultaneously)
- Using general provided development tools to create applications for multiple platforms (e.g.: using only React-Native or Flutter and dart to create a single project for both Android and iOS)
- Might not have as deep access as native app development tools for some specific features or functionalities.
- Should wait until the new features are adopted. (e.g.: new features might not be ready for development with React Native/Flutter)
- Platform Specific features. (Maybe some features are only available for iOS and some for Android)

Flutter Architecture

It functions as a sequence of independent libraries, with each of them depending on the underlying layer. Each part of the framework level is replaceable, and no layer has privileged access to the lower layer. As a rule, developers work with the framework written in the Dart language.





Why Flutter? *(below info has been generated using AI)*

1. **Performance:** Flutter is known for its excellent performance, as it uses its own rendering engine called Skia. This allows Flutter to efficiently render UI elements and deliver smooth animations. React Native, on the other hand, uses a bridge to communicate with native components, which can introduce some performance overhead.
2. **Consistent UI:** Flutter provides a consistent user interface across different platforms. Since Flutter doesn't use native components, it allows developers to build UIs that look and feel the same on both iOS and Android. React Native, on the other hand, uses native components, which can lead to inconsistencies in UI.
3. **Hot Reload:** Flutter has a powerful hot reload feature that allows developers to see changes in the app's UI instantly, making the development process faster and more efficient. React Native also has a similar feature called Fast Refresh, but Flutter's hot reload is often considered to be more reliable and faster.
4. **Programming Language:** Flutter uses Dart, a modern, object-oriented programming language. Dart is known for its simplicity and ease of learning, making it a good choice for developers who are new to mobile app development. React Native, on the other hand, uses JavaScript, which is a widely-used language but can have a steeper learning curve for newcomers.
5. **Widgets:** Flutter has a rich set of customizable widgets that make it easy to build complex UI elements. This eliminates the need for additional libraries or dependencies in many cases. React Native also has a wide range of third-party libraries, but Flutter's built-in widget system makes it easier to create custom UI components.
6. **Growing Community:** While React Native has a more established and larger community, Flutter has been gaining popularity rapidly. This means that there is a growing number of developers and resources available for Flutter development.